MILITARY SPECIFICATION

POWER SUPPLY P-P( )/URC-9

1. SCOPE

1.1 This specification covers Power Supply, P-P( )/URC-9, used to provide for operation of Radio Set AN/URC-9 from a 24 volt d.c. source. P-P( )/URC-9 shall be physically interchangeable with the 115/230 volt 60 c.p.s. power supply, PP-2702/URC-9. The power supply covered by this specification will be procured as a part of Radio Set AN/URC-9Y. The nomenclature AN/URC-9Y indicates that Radio Set AN/URC-9 shall be provided with the 24 volt d.c. plug-in power supply (and without the 115/230 volt 60 c.p.s. power supply).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-901 - Shock Tests, H. I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore; General Specification.
MIL-D-17419 - Drawings, Engineering and Associated Documentation Preparation and Microfilming of (for Electronic Equipment).
MIL-R-22732 - Reliability Requirements for Shipboard and Ground Electronic Equipment.
MIL-R-24182 - Radio Sets AN/SRC-20 and AN/SRC-21 (Including Radio Set AN/URC-9 and Radio Set Control C-3866/SRC).

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N.Y. 10016.)

3. REQUIREMENTS

3.1 General. - The equipments shall be built in accordance with the provisions of MIL-E-16400 to the extent specified herein.

3.2 Reliability. - The following reliability requirements as specified in MIL-R-22732 shall be applicable to this equipment.

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3.2.1 The contractor shall conduct the reliability assurance program that is applicable to a preproduction and production model.

3.2.2 Level. - The reliability goal for this equipment shall be 2000 hours, MTBF.

3.2.3 Reliability evaluation shall be performed in accordance with the procedures for "Reliability Assurance by analysis and prediction" as specified in MIL-R-22732.

3.2.4 Testing. - Quantitative reliability requirements shall be demonstrated as specified herein.

3.3 Design requirements. -

3.3.1 Composition. - The equipment shall consist of the following units:

(a) Power supply, P-P( )/URC-9.
(b) Power cable 20 feet long (with mating connector).

3.3.1.1 Construction (modular). - The requirements for modular construction in MIL-E-16400 are applicable.

3.3.2 Environmental conditions. - The equipment shall be capable of operation under only the following conditions as specified in MIL-E-16400 as modified herein:

(a) Temperature, class 1.
(b) 95 percent humidity.

3.3.3 Electrical design. - The nominal supply voltage shall be 24 volts d.c.

3.3.3.1 Overload protection shall be provided but blown fuse indicators are not required. The type of fuseholders and the type of indicator, if used, shall be approved by the bureau or agency concerned.

3.3.3.2 The power input connector shall be mounted on front of equipment.

3.3.4 Maintenance. - Test points and test features are required in accordance with MIL-E-16400.

3.3.5 Mechanical. -

3.3.5.1 Shock, vibration and inclination. -

3.3.5.1.1 The shock, vibration and inclination requirements of MIL-E-16400 are applicable to this equipment.

3.3.5.1.2 Shock requirements shall be for grade A, class II, medium weight, type A of MIL-S-901.

3.3.5.2 The degree of enclosure shall be splashproof in accordance with MIL-STD-108 when the equipment is plugged into the AN/URC-9.

3.3.6 Controls, indicators and panel layout. -

3.3.6.1 Tactile identification of controls is not required.

3.3.6.2 The following controls shall be classified as shown:

(a) On-off switch; frequently used, no locking device required.
(b) Dimmer control: frequently used, no locking device required.

3.3.6.3 Elapsed time indicators are not required.

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3.4 Performance and detail requirements. -

3.4.1 Equipment. - The equipment shall be a transistorized 24-volt d. c. power supply capable of supplying all the power required by Radio Set AN/URC-9. The equipment shall function with the AN/URC-9 with no modification required to Radio Set AN/URC-9. Rotating type power supplies will not be acceptable.

3.4.2 Operating voltage. - When operated at 23 to 29 volts input, the power supply shall provide normal operation of the AN/URC-9. The filament and high voltage outputs of the power supply shall be regulated to prevent damage to the equipment when the input voltage is raised to 29 volts. When the power supply is operated at 24 volts input the power supply shall provide sufficient voltage in order that the AN/URC-9 radio set shall have an r. f. carrier output of not less than 16 watts at 300 mc. The sensitivity of the radio set, when the d. c. power supply is operated at 24 volts, shall be within 1 db of the sensitivity of the radio set when it is operated from the a. c. power supply.

3.4.3 Input power. - The input power required by the power supply shall be kept to the minimum possible.

3.4.4 Voltages supplied. - The power supply shall provide all a. c. and d. c. voltages required by the URC-9 including 110 volts, 60 c. p. s. for operation of the blower assemblies used in the CY-2959/URC-9 Case, and the RT-581/URC-9 Transceiver.

3.4.5 Grounding. - Neither side of the 24-volt d. c. input power line shall be grounded to the equipment.

3.4.6 Cooling. - The power supply shall employ a cooling system to remove excess heat. The cooling air inlet and outlet for this system shall be located on the front panel of the power supply. A direct current motor shall not be employed in the blower assembly.

3.4.7 Radio frequency filtering. - Particular attention shall be devoted to the prevention of radio frequency interference from the power supply.

3.4.8 Power input connection. - A connector shall be provided on the front panel of the power supply for connection to the 24-volt d. c. power source. Bureau or agency concerned approval is required for the specific connector to be used.

3.4.9 Size. - The power supply shall be of such size and shape that it will be completely physically interchangeable with PP-2702/URC-9.

3.4.10 Weight. - The weight of the power supply (less mating connectors) shall not exceed 75 pounds.

3.4.11 Painting. - The front panel of the power supply shall be primed and painted in accordance with MIL-E-16400. The finish coats shall be baked enamel.

3.4.12 Overload protection. - Overload protection shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments, shall have protective devices, or shall be proportioned to withstand such overload without permanent damage to the equipment. A reversal of input voltage polarity shall not result in damage to the equipment.

3.4.13 Undervoltage protection. - The equipment shall not be damaged by voltages below the low limits specified in 3.4.2 and shall resume normal operation automatically when the voltage returns within the voltage limits specified in 3.4.2.

3.4.14 Overvoltage protection. - Means shall be provided to prevent damage to the equipment caused by input voltage transients up to 32 volts which have a time duration up to 0.05 second, at repetition rates of not less than 1 per second, and not more than 30 pulses per second.

3.4.15 Duty cycle. - The power supply shall be capable of continuous duty when the radio set is in transmit operation.

3.4.16 Front panel layout. - The front panel of the power supply shall have a power input connector, on-off switch, fuse holders, pilot light, a dimmer control, and shall contain spare fuses and pilot bulbs.

3.4.17 Instruction plates. - An instruction plate shall be provided on the front panel which shows power requirements, power input connections and any precautions which need to be observed.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 The P-P( )/URC-9 will be procured as a part of Radio Set AN/URC-9Y (24 volt d.c. version of AN/URC-9) and the quality assurance provisions shall be as specified for the AN/URC-9 in MIL-R-24182.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. -

5.1.1 Basic equipment or item. -

5.1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts. - Preparation for delivery requirements for onboard repair parts shall be as specified in the contract or order for the repair parts.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation, packaging, packing and marking. - The equipment, accessories, repair parts and manuals shall be preserved and packaged level A or C; packed level A, B or C; and marked as specified in accordance with MIL-E-17555.)

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Preservation, packaging, packing or marking requirements other than those required by 5.1 (see 5.2).

Preparing activity:
Navy - SH
(Project 5820-N477Sh)

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